higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

INTRODUCTION TO SYSTEMS DEVELOPMENT

(First Paper)

NQF LEVEL 2

(10041022)

12 March 2019 (X-Paper)
09:00–11:00

This question paper consists of 7 pages.

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TIME: 2 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Which ONE of the following is an end-user program:

A Debugger
B Compiler
C Word processor
D Translator

1.2 Which ONE of the following is NOT a high-level programming language:

A Java
B Assembly
C Cobol
D Python

1.3 … interfaces are often difficult to use because they require exact spelling, grammar and punctuation.

A Graphical user
B Command-line
C Menu-driven
D Icon-driven

1.4 Which operating system would normally be installed on a desktop PC at home?

A Windows 7
B Windows Embedded Compact
C Windows Server 2008
D Windows Mobile

1.5 The default data type of content inside a textbox (TextBox.Text):

A Byte
B Bit
C String
D Integer

1.6 Which ONE of the following is an example of a first-generation programming language:

A Fortran
B Machine code
C Java
D Cobol

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1.7 Operating system used on big network systems:

A UNIX
B Linux
C MS-DOS
D OS2

1.8 Which operating system was the first system to use the GUI environment?

A Windows
B Apple OS
C Linux
D UNIX

1.9 Which ONE of the following is a string:

A "7523061"
B 23.10
C True
D 45

1.10 Which ONE of the following is NOT an example of a database-management system?

A SQL Server 2008
B Oracle 11g
C Windows Server 2008
D IBM Informix Dynamic Server

(10 × 1) **[10]**

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.1–2.10) in the ANSWER BOOK.

2.1 A user does not need access to a working internet connection to get online help.

2.2 An algorithm is a procedure or formula for accomplishing a task.

2.3 The main benefit of low-level languages is flexibility.

2.4 The ability to prevent future system failures and avoid problems before they occur is called change management.

2.5 A record is a collection of information organised in a way which facilitates quick access to information.

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2.6 Methods to gather information include questionnaires and interviewing clients.

2.7 Defensive programming is a way of writing source code so that the program can continue working in spite of unforeseen actions and inputs by the user.

2.8 Pseudocode is a generic, compact, informal way of writing an algorithm using the conventions of programming languages, but without language-specific syntax.

2.9 A software bug is a feature that ensures that a computer program works as intended and produces a correct result.

2.10 The maintenance of software incorporates all the development phases of the software product.

(10 × 1) **[10]**

QUESTION 3

3.1 Name the FOUR fundamental features of application software. (4)

3.2 Give TWO examples of language translators. (2)

3.3 One of the reasons why software applications share common features is for the sake of compatibility.

Explain *backward compatibility* and *forward compatibility*. (2 + 2) (4)

3.4 List FIVE characteristics of operating systems. (5)

3.5 Device drivers can be categorised as system software.

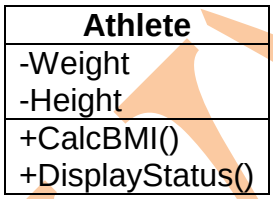
What is the purpose of device drivers? (1)
[16]

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QUESTION 4

4.1 Give FOUR advantages and FOUR disadvantages of high-level programming languages. (4 + 4) (8)

4.2 Study the class diagram below and answer the questions.



4.2.1 The name of the given class is Athlete.

What is a *class*? (3)

4.2.2 Briefly differentiate between *attributes* and *methods*. (2)

4.2.3 Give TWO attributes of the given class. (2)

4.2.4 Give TWO methods of the given class. (2)

4.3 The following questions concern the programming language Visual Basic.NET and its concepts.

4.3.1 What are the benefits of code recycling? (5)

4.3.2 List THREE object-oriented programming languages, apart from Visual Basic.NET. (3)

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QUESTION 5

5.1 Every programming team has to follow certain basic steps when developing new software.

List FIVE steps involved in the program-development life cycle (PDLC). (5)

5.2 What is the EBCDIC coding system used for? (1)

5.3 Evaluate the expression 5 – 3 ^ 2 \ 8 + 17 mod 3 * 2 (show ALL steps). (5)

[11]

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QUESTION 6

6.1 The process of finding errors is called debugging.

Name FOUR categories of errors. (4)

6.2 List FOUR common toolbox controls used when developing a program in Visual Basic.NET. (4)

6.3 The following marks were achieved by students writing a programming test:

31, 35, 39, 42, 45, 49, 52, 56, 57, 57, 62, 65, 78, 79

Use this dataset to answer the following questions:

6.3.1 Draw a frequency-distribution table, using mark ranges 30-39; 40-49; 50-59; 60-69; and 70-79, with the following table headings:

Class (Mark range)	**Number of students**

(5)

6.3.2 Use the frequency-distribution table drawn in QUESTION 6.3.1 to draw a bar chart. Use 'Number of students' for the vertical axis and 'Mark range' for the horizontal axis. (5)

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QUESTION 7

7.1 Name and explain TWO testing methods for verification. (2 × 2) (4)

7.2 Differentiate between *feasibility* and *viability*. (2 + 2) (4)

7.3 Give TWO advantages of having a QA team involved while developing a software product. (2)

[10]

TOTAL: 100

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